

0 5 . 1

A student prepared a stained squash of cells from the tip of an onion root and observed it using an optical microscope.

During the preparation of the slide, he:

- cut the first 5 mm from the tip of an onion root and placed it on a glass slide
- covered this tip with a drop of stain solution and a cover slip
- warmed the glass slide
- pressed down firmly on the cover slip.

He identified and counted nuclei in different stages of the cell cycle.

Explain why the student:

[2 marks]

1. used only the first 5 mm from the tip of an onion root.

2. pressed down firmly on the cover slip.

Question 5 continues on the next page

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Figure 2 shows the cells the student saw in one field of view. He used this field of view to calculate the length of time these onion cells spent in anaphase of mitosis.

Figure 2



0 5 . 2

Scientists have found the mean length of time spent by onion cells in anaphase of mitosis is 105 minutes. They also found the cell cycle of cells in the onion root shown in **Figure 2** takes 1080 minutes.

32 whole cells are shown in **Figure 2**.

Use this information and **Figure 2** to calculate the length of time the cells of this onion root are in anaphase **and** then calculate the percentage difference between your answer and the mean length of time found by the **scientists**.

Show your working.

[2 marks]

Answer = _____ %



0 5 . 3

Tick (✓) the name given to the division of cytoplasm during the cell cycle.

[1 mark]

Binary fission

☐

Cytokinesis

☐

Phagocytosis

☐

Segregation

☐**0 5 . 4**

Describe and explain what the student should have done when counting cells to make sure that the mitotic index he obtained for this root tip was accurate.

[2 marks]

Question 5 continues on the next page**Turn over ►**

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A scientist treated growing tips of onion roots with a chemical that stops roots growing. After 24 hours, he prepared a stained squash of these root tips.

Figure 3 is a drawing showing the chromosomes in a single cell observed in the squash of one of these root tips in anaphase. This cell was typical of other cells in anaphase in these root tips.

Figure 3



Use all of this information to suggest how the chemical stops the growth of roots.

[3 marks]

[illegible]

10

